

① I)

$$i_t = i_t^* + E_t[\Delta S_{t+1}]$$

$$\Delta S_{t+1} = S_{t+1} - S_t$$

$$i_t = i_t^* + E_t[S_{t+1} - S_t]$$

$$i_t = i_t^* + E_t[S_{t+1}] - E_t[S_t] \quad \leftarrow E_t[S_t] = S_t$$

$$i_t = i_t^* + E_t[S_{t+1}] - S_t$$

$$S_t = \cancel{(i_t^* - i_t)} + E_t[S_{t+1}] + (i_t^* - i_t)$$

so:

$$S_{t+1} = E_{t+1}[S_{t+2}] + (i_{t+1}^* - i_{t+1})$$

this is equal to $E_t(i_t^* - i_t)$

Substituting that in, and then S_{t+2} etc. until ∞ gives:

$$\begin{aligned} S_t &= E_t[S_{t+\infty} + (i_t^* - i_t) + (i_{t+1}^* - i_{t+1}) + (i_{t+2}^* - i_{t+2}) \dots] \\ &= E_t(S_{t+\infty}) + E_t\left[\sum_{j=0}^{\infty} (i_{t+j}^* - i_{t+j})\right] \end{aligned}$$

II)

PPP:

$$\frac{p^* \bar{s}}{p} = 1$$

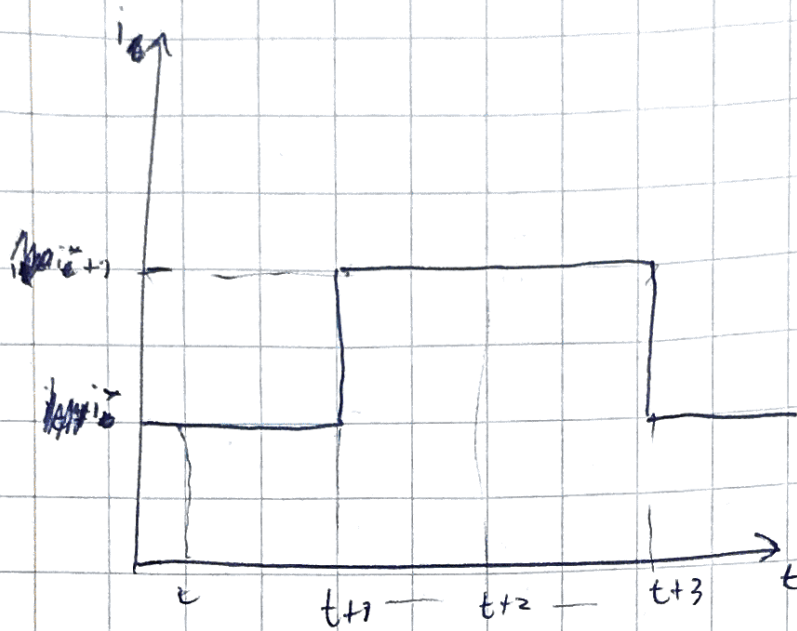
$$\bar{s} = \frac{p}{p^*}$$

This is if you're using big (capital) S
taking the $\log(S) = s$
the ppp is:

$$q = p^* + s - p = 0$$

$$\Rightarrow s = p - p^*$$

III)

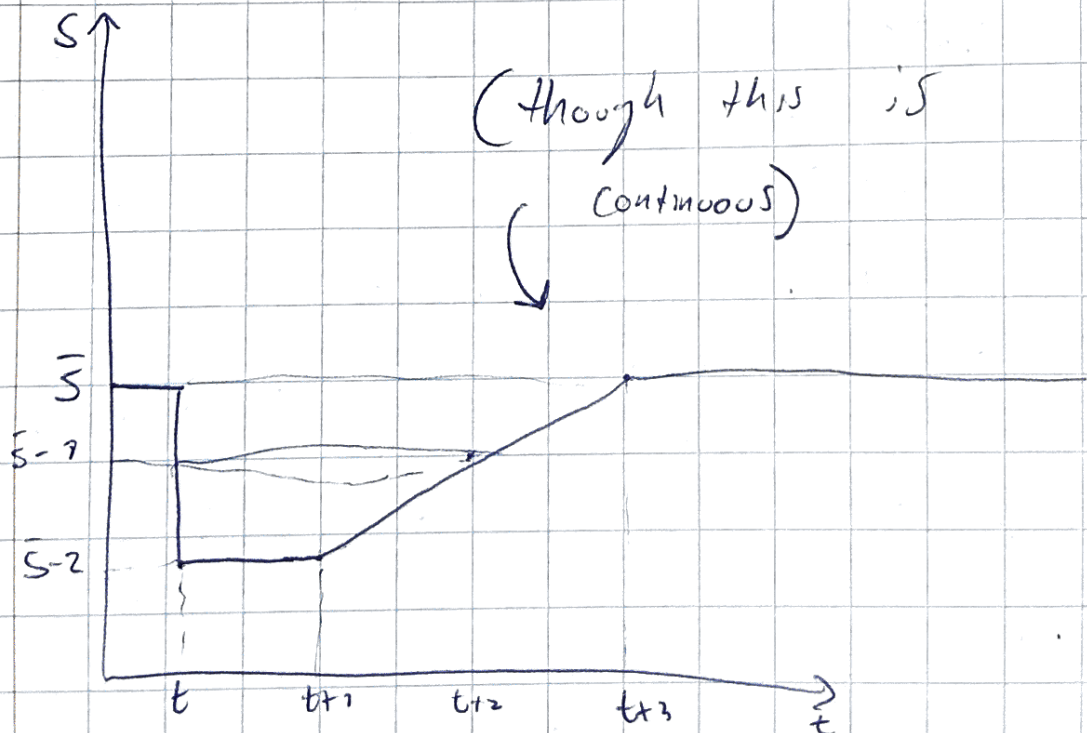


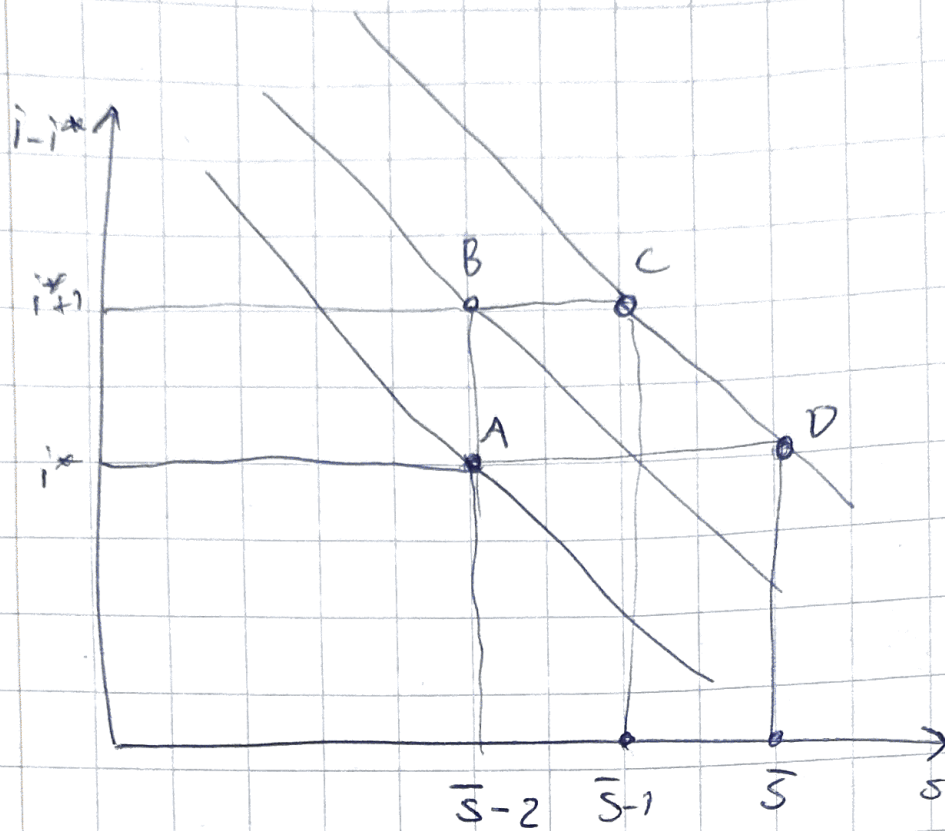
$$S_0 = \bar{S} + 0 - 1 - 1 + 0 = -2$$

$$S_{t+1} = \bar{S} + -1 - 1 + 0 = -2$$

$$S_{t+2} = \bar{S} + -1 + 0 = -1$$

$$S_{t+3} = \bar{S} + 0 = 0$$





But probably better not
on the same graph